

WATER & ECOSYSTEM

Connected Waters

One-line value proposition:

A field-based programme connecting freshwater ecosystems, wetlands, biodiversity and water stewardship through practical environmental learning.

1. Project Title

Water & Ecosystem — Connected Waters

2. Context / Problem

Freshwater ecosystems connect rivers, wetlands, surrounding land, wildlife and people. These relationships can be difficult to understand without seeing them directly.

HWK already uses wetlands, waterbirds, aquatic biodiversity and the Eurasian beaver as practical entry points for environmental learning.

Existing activities introduce participants to wetland ecosystems, aquatic invertebrates, migratory birds, river biodiversity, beaver ecology, water pollution and the relationship between freshwater habitats and surrounding landscapes.

These activities provide a foundation for a broader Water & Ecosystem field-learning programme.

3. Project Objective

To strengthen practical understanding of freshwater ecosystems and water stewardship through field observation, environmental education and collaboration with relevant specialists.

4. Target Group

- Young people and students
- Schools and education groups
- Community participants
- Educators
- Participants interested in water, biodiversity and ecosystem learning

Participant numbers and age range: [TO CONFIRM]

5. Proposed Activities

1. **River and wetland field-learning sessions** exploring connections between water, land and biodiversity.
2. **Freshwater biodiversity observation**, including selected aquatic life, waterbirds and wetland habitats.
3. **Beaver and river-ecosystem learning**, where appropriate, using beaver ecology to explain wetland formation, water retention and habitat relationships.
4. **Migratory bird and wetland learning** focused on the importance of feeding, resting and breeding habitats.
5. **Basic field observation or sampling demonstrations** led by appropriately qualified specialists, including aquatic invertebrate learning where relevant.
6. **Water stewardship learning**, including discussion of human impacts, water pollution, ecosystem protection and responsible use of freshwater environments.

6. Expected Outputs

- A structured Water & Ecosystem field-learning programme.
- River and wetland learning modules.
- Expert-supported field activities.
- Practical freshwater biodiversity observation sessions.
- Educational resources connecting water, biodiversity and land.
- Opportunities for partner-led research, monitoring or knowledge-exchange activities.

7. Expected Outcomes

The programme is intended to:

- improve understanding of freshwater and wetland ecosystems;
- strengthen awareness of relationships between water, biodiversity and land;
- build practical knowledge through field observation;
- increase understanding of human pressures on freshwater systems;
- strengthen water stewardship awareness;
- create opportunities for collaboration between learners, educators, researchers and ecosystem specialists.

8. Expected Impact

The long-term intended contribution is stronger environmental understanding and stewardship of freshwater ecosystems.

The programme aims to help participants see rivers and wetlands as connected ecological systems rather than isolated water bodies, and to understand why protecting freshwater habitats also matters for biodiversity and surrounding landscapes.

HWK does **not** claim that this educational programme itself will deliver quantified ecosystem restoration outcomes.

9. HWK Approach

The programme applies:

Hike → Learn → Experience → Care → Protect

Hike — enter river, wetland and related landscapes.

Learn — understand freshwater ecology with relevant experts and educators.

Experience — observe aquatic life, waterbirds, habitats and ecological relationships directly.

Care — build appreciation for freshwater systems and their role in wider landscapes.

Protect — connect knowledge with water stewardship and responsible environmental action.

10. Partnership Opportunity

Potential partner contributions include:

- programme funding or sponsorship;
- freshwater and wetland expertise;
- biodiversity research;
- migratory-bird and flyway expertise;
- training and technical capacity;
- field equipment and educational resources;
- research or monitoring collaboration;
- institutional partnerships;
- joint field programmes;
- knowledge exchange.

Research and monitoring components would be developed with appropriate technical partners rather than presented as existing HWK research capacity.

11. Funding Need

Funding amount: [TO CONFIRM]

Potential resource categories include:

- programme coordination;
- field transport and logistics;
- expert facilitation;
- freshwater and biodiversity observation equipment;
- educational materials;
- field documentation;
- partner-led technical training;
- agreed research or monitoring activities.

Exact funding requirements will depend on confirmed scope, sites, duration and partner responsibilities.

12. Scalability / Replication Potential

The concept can be developed through complementary modules:

River Ecosystems → Wetlands → Beaver Ecology → Aquatic Biodiversity → Waterbirds & Flyways → Water Stewardship

This modular structure allows the programme to be adapted to different freshwater themes, learning groups and institutional interests.

Future implementation locations beyond existing HWK activity settings are **[TO CONFIRM]**.

13. Why HWK

HWK already delivers field learning connected to:

- wetland ecosystems;
- waterbirds;
- aquatic invertebrates;
- the Tuul River ecosystem;
- Eurasian beaver ecology;
- freshwater biodiversity;
- water pollution and conservation awareness.

HWK's role is to make environmental relationships visible and understandable through field experience while creating a platform for scientists, educators, communities and young people to learn together.

14. Suggested Partner Types

- Water and wetland organizations
- River and freshwater initiatives
- Biodiversity and conservation NGOs
- Universities and research institutions
- Migratory-bird and flyway organizations
- Youth and environmental education programmes
- Environmental foundations and grant-makers
- Corporate / ESG partners focused on water or biodiversity

Partnership Ask

Partner with Hiking with Knowledge to co-create field learning that makes the connections between water, biodiversity, land and people visible, understandable and worth protecting.